6 The nth term of a sequence is $2n^2 - 1$

The nth term of a different sequence is $40 - n^2$

Show that there is only one number that is in both of these sequences.

For n = 1,2,3,4,5,6:

$2n^2 - 1$: 1, 7, 17, 31, 49, 71

$40 - n^2$: 39, 36, 31, 24, 15, 4

The only common term is 31 (at n = 4 and 3).

The first grows; the second falls, so none later.

3 marks total
Find terms of both
Continue beyond 31
Only one common: 31

(Total for Question 6 is 3 marks)